

Vehicle Number Plate Detection

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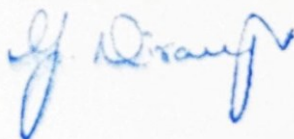
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ABSTRACT

Number Plate recognition, also called License Plate realization or recognition using image processing methods, is a potential research area in smart cities and the Internet of Things. An exponential increase in the number of vehicles necessitates the use of automated systems to maintain vehicle information for various purposes.

In the proposed algorithm, an efficient method for recognizing Indian vehicle number plates has been devised. This method is able to handle noisy, low illuminated, cross-angled, and non-standard font number plates. This work employs several image processing techniques in the preprocessing stage, including morphological transformation, Gaussian smoothing, Gaussian thresholding, and the Sobel edge detection method.

After preprocessing, the algorithm applies number plate segmentation. Contours are determined by border following, and these contours are filtered based on character dimensions and spatial localization. Finally, Optical Character Recognition (OCR) is applied to recognize the extracted characters. The detected texts are stored in a database, allowing for sorting and search functionality.

While this project demonstrates effective results, it has certain drawbacks and limitations due to the absence of advanced machine learning or deep learning algorithms. However, it works efficiently for an average use case.

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ABBREVIATIONS

The following abbreviations are used in this project:

- **AI** – Artificial Intelligence
- **CNN** – Convolutional Neural Network
- **OCR** – Optical Character Recognition
- **ML** – Machine Learning
- **ROI** – Region of Interest (specific area for detection/recognition)
- **TP** – True Positive
- **FP** – False Positive
- **FN** – False Negative
- **TN** – True Negative
- **MAP** – Mean Average Precision (a performance metric for object detection)
- **LPR** – License Plate Recognition

CHAPTER 1

INTRODUCTION

Number Plate Recognition (NPR) is a crucial component in intelligent transportation systems, particularly in applications like traffic management, toll collection, and security. In a rapidly growing urban environment, where the number of vehicles is increasing exponentially, there is a significant need for automated systems to monitor and track vehicles effectively. In this project, we have developed a Number Plate Recognition system specifically tailored for Indian vehicles, leveraging OpenCV for image processing, Convolutional Neural Network (CNN) for training the model and Tesseract for Optical Character Recognition (OCR). This system processes images to detect and recognize license plate characters, even under challenging conditions, such as low light, noisy backgrounds, and non-standard fonts.

1.1 Problem Statement

The growing number of vehicles on the road has led to an increased demand for automated systems capable of recognizing and managing vehicle-related data. Traditional manual methods of vehicle identification, such as using human observation or entering license plate numbers manually, are becoming inefficient, error-prone, and impractical. This has highlighted the need for automated License Plate Recognition (LPR) systems that can quickly and accurately identify vehicles in real-time, without human intervention.

In particular, recognizing Indian vehicle number plates presents unique challenges due to variations in fonts, sizes, backgrounds, and the presence of environmental factors such as low light or obstructions. The main issues include:

- **Noisy Images:** Images may have poor quality due to weather conditions, camera quality, or lighting, making it difficult to distinguish number plates from the rest of the image.
- **Angled or Distorted Plates:** Plates may be viewed at an angle or have distortions due to the vehicle's movement.
- **Non-standard Fonts:** Indian number plates often use non-standard fonts that can vary across different states or regions.

The goal of this project is to overcome these challenges and develop an efficient system that can automatically detect and recognize vehicle number plates, specifically for Indian vehicles, in varying environmental conditions. This system should employ image processing techniques, CNN and Optical Character Recognition (OCR) to accurately identify the text on the plates and store the results for further processing, such as in traffic management or law enforcement.

1.2 OBJECTIVES

The objectives of this project are as follows:

1. Develop an automated system for Number Plate Recognition (NPR) that can detect and recognize vehicle number plates with high accuracy, even under adverse conditions such as low light or unclear images.
2. Handle real-world challenges such as noisy, distorted, or angled number plates, common in practical applications like traffic monitoring or security surveillance.
3. Use image processing techniques to preprocess images, including morphological transformations, Gaussian smoothing, and edge detection, to improve the quality of the image and facilitate accurate plate detection.
4. Implement Optical Character Recognition (OCR) using Tesseract or another suitable OCR engine to extract the characters from the segmented number plates and store the recognized information in a database for further use.
5. Build a user-friendly system that can be integrated into existing traffic management systems or parking applications, allowing for efficient and real-time recognition of vehicle plates.
6. Evaluate the system's performance in terms of accuracy, speed, and reliability, comparing it with other existing systems or approaches to highlight strengths and limitations.

1.3 Software Requirements

The software for this project is designed to perform number plate detection and recognition using OpenCV and Tesseract OCR. Below is the specification of the required software:

➤ Functional Requirements

1. Image Preprocessing Module:
 - Apply Gaussian smoothing to reduce image noise.
 - Use edge detection techniques (Sobel, Canny) for plate localization.
 - Apply thresholding for binary conversion of images to enhance contrast.
2. Plate Detection:
 - Detect the region of interest (ROI) that corresponds to the number plate using contour detection and border-following techniques.

- Handle different plate orientations and partial obstructions.
- 3. Character Segmentation:
 - Segment the detected number plate into individual characters for OCR.
- 4. Optical Character Recognition (OCR):
 - Use Tesseract OCR to recognize characters from the segmented image of the number plate.
- 5. Database Integration:
 - Store recognized number plates in a database for further querying or logging.
- 6. Performance Monitoring:
 - Real-time monitoring of system performance with accuracy metrics and potential error logs.
- 7. Convolutional Neural Network (CNN):
 - Generate Synthetic Data for Training the CNN Model which is used to for Character Recognition.

➤ **Non-Functional Requirements**

1. Platform:
 - The system should be compatible with both Windows platforms.
2. Speed:
 - The system should process and recognize a number plate in less than 5 seconds for real-time applications.
3. Scalability:
 - The system should be scalable to handle large datasets, with the ability to add more vehicle data to the database for retrieval.
4. Usability:
 - The system should have a simple user interface (UI) that allows for easy interaction with the detection and recognition process.

➤ **Software Tools and Libraries**

1. OpenCV:

- A comprehensive computer vision library used for image processing tasks such as edge detection, morphological operations, and contour detection.
- 2. Tesseract OCR:
 - An open-source OCR engine that will be used for extracting text from images.
- 3. Python:
 - The primary programming language for implementing the system, due to its support for both OpenCV and Tesseract OCR.
- 4. Integrated Development Environment (IDE):
 - Visual Studio Code (VS Code) or PyCharm for coding and debugging.
- 5. Convolutional Neural Network (CNN):
 - Generate Synthetic Data for Training the CNN Model which is used to for Character Recognition.

1.4 Scope of the Project

The Number Plate Recognition (NPR) system is designed to automate the identification of vehicle license plates from images or video streams. This system's scope spans several applications in public safety, transportation management, and access control. Here's a detailed breakdown of its scope:

1. Traffic Law Enforcement

- **Automated Violation Detection:** The NPR system can be deployed at intersections, highways, and other locations to monitor vehicle compliance with traffic rules. It can identify vehicles involved in offenses such as speeding, red-light violations, and unauthorized lane usage.
- **Real-time Reporting:** Once a violation is detected, the system can instantly report the license plate number to a central authority, providing evidence that can be used for issuing fines or warnings.
- **Reduction of Human Intervention:** By automating vehicle monitoring and plate recognition, the system reduces the need for manual checks and human intervention, allowing traffic police to focus on critical tasks.

2. Parking Management

- **Automated Entry and Exit Logging:** The system can be used at parking facilities to log the entry and exit of vehicles. This eliminates the need for physical tickets, as vehicles are automatically identified by their license plates.
- **Enhanced Security:** The system enables parking management to maintain a record of all vehicles, enhancing security by allowing only authorized vehicles into the facility.

- **Fee Calculation:** With timestamps associated with entry and exit logs, the system can calculate parking fees based on the duration of the vehicle's stay, making the payment process faster and more convenient.

3. Toll Collection

- **Faster Toll Processing:** At toll booths, the NPR system can automatically identify vehicle plates, allowing for quicker toll collection. This reduces traffic congestion and improves the flow of vehicles at toll stations.
- **Seamless Integration with Electronic Payment Systems:** NPR systems can work with digital payment methods, like RFID-based toll collection, where vehicles are identified by both their license plates and electronic tags. This creates a robust, error-free toll collection method.
- **Cost Reduction:** By automating toll collection, the system reduces the need for manual toll booth operators, thereby lowering operational costs.

4. Access Control for Restricted Areas

- **Gated Communities and Corporate Campuses:** The NPR system can control vehicle access by checking license plates against a list of authorized vehicles, enhancing security for residential or commercial areas.
- **Government Facilities and High-Security Zones:** In sensitive areas, such as government or military facilities, the system can prevent unauthorized vehicles from entering by granting access only to recognized license plates.
- **Event and Conference Management:** During events, NPR systems can manage parking and access for attendees, allowing only registered vehicles to enter designated parking areas.

5. Public Transport and Fleet Management

- **Fleet Tracking and Management:** Organizations with a large fleet of vehicles, such as logistics companies, can use NPR systems to track vehicle locations and log entry/exit times, ensuring efficient management and scheduling.
- **Vehicle Identification in Public Transport Systems:** NPR systems can aid in identifying public transportation vehicles and monitoring their routes, schedules, and adherence to designated stops.
- **Enhanced Safety and Accountability:** The system can log vehicle information, providing a record that can be used to investigate incidents or track vehicle usage.

CHAPTER 2

LITERATURE SURVEY

This literature survey presents an overview of significant studies and techniques in the domain of Number Plate Detection, focusing on methodologies, innovations, and limitations encountered in various approaches. The section is organized into Existing Systems and Approaches and Limitations of Existing Systems to highlight progress and remaining challenges in the field.

2.1 Existing Systems and Approaches

1. "Number Plate Recognition for Indian Vehicles" – M. M. Shidore and S. P. Narote (2011)"

This paper presents a method for recognizing Indian vehicle license plates using traditional image processing techniques. The system employs edge detection, morphological operations, and Optical Character Recognition (OCR) for number plate detection and character recognition. The approach is designed to handle noisy images, low illumination, and cross-angled plates. However, the system's performance was limited when dealing with variations in plate fonts and poor image quality, which is typical in real-world environments.

2. "A Recognition of Vehicle License Plate Using a Genetic Algorithm-Based Segmentation" – Sang Kyoon Kim, D. W. Kim, and Hang Joon Kim (1996)"

This research focuses on the use of genetic algorithms for license plate segmentation. The algorithm is designed to optimize the segmentation process, allowing for more accurate extraction of the license plate from an image, even when the plate is tilted or positioned at unusual angles. The authors demonstrate that the genetic algorithm improves segmentation accuracy by adapting to various plate positions and sizes. However, the method still faces challenges in real-time applications due to the computational complexity of the genetic algorithm, especially in diverse environments with varying lighting conditions.

3. "Comparative Analysis of License Plate Recognition Systems" – Bai et al. (2022)"

Bai et al. explore the use of machine learning algorithms like Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) for character recognition in license plate recognition systems. In addition, Haar Cascade classifiers were used for plate detection. The system showed improvements over traditional methods, providing better performance in terms of speed and accuracy in detecting plates from different angles. However, the study also highlights that these systems still face challenges with complex environmental factors, such as non-standard plate fonts and poor lighting.

4. "End-to-End License Plate Recognition Using Deep Learning" – Huang et al. (2018)"

Huang et al. utilized Convolutional Neural Networks (CNNs) for license plate detection and Recurrent Neural Networks (RNNs) for character recognition, forming an end-to-end deep learning-based license plate recognition system. This system improved accuracy and robustness across a range of environmental conditions, including variations in plate designs, lighting, and noise. The deep learning model was able to detect and recognize characters efficiently. However, the system requires large labelled datasets for training and is computationally intensive, making it less feasible for real-time applications without high-end hardware.

5. "Challenges in License Plate Recognition: An Investigation into Real-World Applications" – Various Authors (2022)"

This paper provides an overview of the challenges faced by existing LPR systems when applied to real-world scenarios. Factors such as plate orientation, varying plate designs, and complex environmental conditions (e.g., poor lighting, reflections, and occlusions) are discussed in detail. The paper highlights that while machine learning and deep learning models have made significant strides in LPR, they still struggle with generalization across different countries and plate designs. Real-time processing and the need for large training datasets also present challenges for large-scale deployment.

2.2 Limitations of Existing Systems

1. "Number Plate Recognition for Indian Vehicles" – M. M. Shidore and S. P. Narote (2011)"

Despite its effectiveness in controlled conditions, this system's performance significantly degrades when applied to images with noise, poor lighting, or non-standard fonts. This limitation makes it unsuitable for real-world applications where environmental conditions often vary, requiring more robust methods to handle these complexities.

2. "A Recognition of Vehicle License Plate Using a Genetic Algorithm-Based Segmentation" – Sang Kyoon Kim, D. W. Kim, and Hang Joon Kim (1996)"

While the use of genetic algorithms for segmentation improved accuracy, the approach suffered from high computational costs, limiting its real-time application. Moreover, the method struggled with diverse plate designs and complex environmental conditions, such as reflections and shadows, which are common in practical scenarios.

3. "Comparative Analysis of License Plate Recognition Systems" – Bai et al. (2022)"

Though SVM and KNN algorithms provided improved accuracy over traditional methods, the study identifies limitations in dealing with varying font styles, plate orientations, and environmental challenges. The need for manual tuning of parameters in SVM-based systems and the inability of KNN to handle large datasets efficiently are significant drawbacks.

4. "End-to-End License Plate Recognition Using Deep Learning" – Huang et al. (2018)"

The deep learning approach significantly improved accuracy but has several limitations. These include the need for large, labeled datasets for training and the heavy computational requirements, which make the system less feasible for real-time applications. Moreover, the system can still struggle with uncommon plate designs or damage to the plates, impacting performance.

5. "Challenges in License Plate Recognition: An Investigation into Real-World Applications" – Various Authors (2022)"

The paper emphasizes the limitations of current LPR systems in generalizing across diverse plate designs and environmental conditions. Factors such as plate damage, occlusion, poor lighting, and varying plate fonts hinder the reliability of these systems. Additionally, real-time processing remains a challenge due to the computational load required by advanced machine learning and deep learning models.

2.3 Comparative Analysis of Techniques

Various techniques have been explored in the field of Number Plate Recognition (NPR), each with distinct approaches, benefits, and limitations.

Technique	Description	Advantages	Limitations	Use Cases
Optical Character Recognition (OCR)	Traditional approach using OCR algorithms on pre-processed images to extract plate characters.	High accuracy in controlled environments with clear, standard-format plates.	Limited performance in noisy, low-quality images or with non-standard fonts.	Parking management, toll collection.
Edge Detection (e.g., Sobel, Canny)	Identifies plate region by detecting high-contrast edges in the image.	Effective in locating plates in clear, well-contrasted images.	Sensitive to noise, lighting, and complex backgrounds.	Plate localization in controlled settings.
Convolutional Neural Networks (CNNs)	Leverages CNNs for detecting and classifying characters in images.	High accuracy, adaptable to noise and diverse fonts, handles complex backgrounds.	Requires a large dataset for training and high computational resources.	Traffic enforcement, security.
Template Matching	Matches characters against pre-defined templates to recognize text on plates.	Simple to implement, effective for uniform and structured plates.	Fails with variable fonts, noisy images, and non-standard plate formats.	Fixed-format environments.
Genetic Algorithms	Uses genetic algorithms to optimize segmentation and character recognition.	Robust in segmentation, adaptable to varying conditions.	Computationally intensive, slower than traditional methods.	Complex environments, experimental settings.

Table 2.1: Comparison of Techniques Used

CHAPTER 3

METHODOLOGY OF NUMBER PLATE SYSTEM

The methodology for the Number Plate Recognition (NPR) system is designed to identify and read the vehicle license plate from an image using image processing techniques. The overall approach involves multiple stages starting from image acquisition to recognition and display.

3.1 Image Preprocessing

Image preprocessing is a crucial step to enhance the image quality and prepare it for further analysis. The purpose of this step is to improve the image, eliminate noise, and highlight essential features, making it easier to detect the number plate in the subsequent steps.

3.1.1 Noise Reduction

In real-world scenarios, images of vehicles can be noisy due to factors like poor lighting, camera quality, or interference from the environment (e.g., reflections or weather conditions). The noise in an image can severely hinder the recognition process. Therefore, noise reduction is applied using Gaussian smoothing.

- **Gaussian Smoothing:** This technique smoothens the image by applying a Gaussian filter that reduces the noise without blurring the important edges. The filter works by averaging the pixel values in a neighbourhood around each pixel. A Gaussian kernel is used for convolution, which applies a weighted average based on the distance from the center pixel.

Mathematically, the Gaussian filter is represented as:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

where σ is the standard deviation, determining the extent of smoothing.

- **Implementation:** In OpenCV, the function `cv2.GaussianBlur()` is used to apply this technique, reducing the high-frequency noise present in the image.

3.1.2 Edge Detection and Segmentation

After noise reduction, the next step is to detect the edges of the image and extract the relevant regions (i.e., the license plate) using edge detection methods. This is crucial for image segmentation where the goal is to isolate the number plate from the rest of the vehicle image.

- **Sobel Edge Detection:** Sobel edge detection is a technique to identify changes in intensity and highlight the edges in an image. It is implemented using two kernels: one for detecting horizontal edges and the other for detecting vertical edges.

The Sobel operator for horizontal edges (G_x) and vertical edges (G_y) is:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

- **Segmentation:** After edge detection, the system performs contour detection using `cv2.findContours()`. This step identifies continuous regions in the image and locates potential license plates. The contours are filtered based on their size, shape, and position in the image to eliminate irrelevant objects.

3.2 License Plate Localization

This step focuses on isolating the region containing the license plate:

- **Segmentation Methods**

Segmentation separates the number plate from the background to narrow down the area for character detection.

- **Thresholding:** Sets a threshold to isolate bright areas from the darker background, useful when the number plate contrasts sharply with its surroundings.
- **Morphological Operations:** Techniques like dilation and erosion remove small imperfections and enhance the plate area by adjusting pixel connectivity.

- **Contour Detection**

Contours represent the boundaries of detected shapes within the image, helping identify potential number plate areas.

- **Finding Contours:** Using algorithms to trace the boundaries of high-intensity areas, contour detection can isolate rectangular shapes that may represent a plate.
- **Filtering by Aspect Ratio:** Only contours with an aspect ratio similar to a standard license plate are selected, improving localization accuracy.

3.3 Character Segmentation

Once the license plate is located, the next step is to segment individual characters:

- **Character Isolation**

Characters on the plate are separated to allow individual recognition.

- **Vertical Projection Method:** Uses pixel density along vertical lines to detect spaces between characters.
- **Connected Component Labelling:** Labels connected pixel regions, isolating each character within the plate.

- **Handling Non-Standard Fonts**

Number plates may contain varying fonts and character sizes, posing a challenge for recognition.

- **Font Variability Adaptation:** Configures OCR parameters to accommodate variations in font style, size, and alignment.
- **Aspect Ratio Analysis:** Characters are filtered by aspect ratio to reduce errors due to distorted or non-standard fonts.

3.4 Optical Character Recognition (OCR)

OCR converts the segmented characters into machine-readable text, identifying the letters and digits on the number plate.

- **Tesseract OCR Implementation**

Tesseract is a popular open-source OCR engine that can be configured to recognize alphanumeric characters on number plates.

- **Training for Alphanumeric Characters:** Tesseract is trained to focus on standard characters typically found on number plates.
- **Enhancement Techniques:** Adjusts parameters to optimize OCR accuracy, especially in scenarios with varied lighting or character distortion.

- **Error Correction Techniques**

To improve accuracy, post-processing methods are applied to correct OCR errors.

- **Dictionary-Based Correction:** A pre-defined dictionary of common license plate formats can be used to correct misrecognized characters.

- Pattern Matching: Corrects errors by comparing recognized characters against known patterns (e.g., letters followed by numbers).

3.5 System Architecture

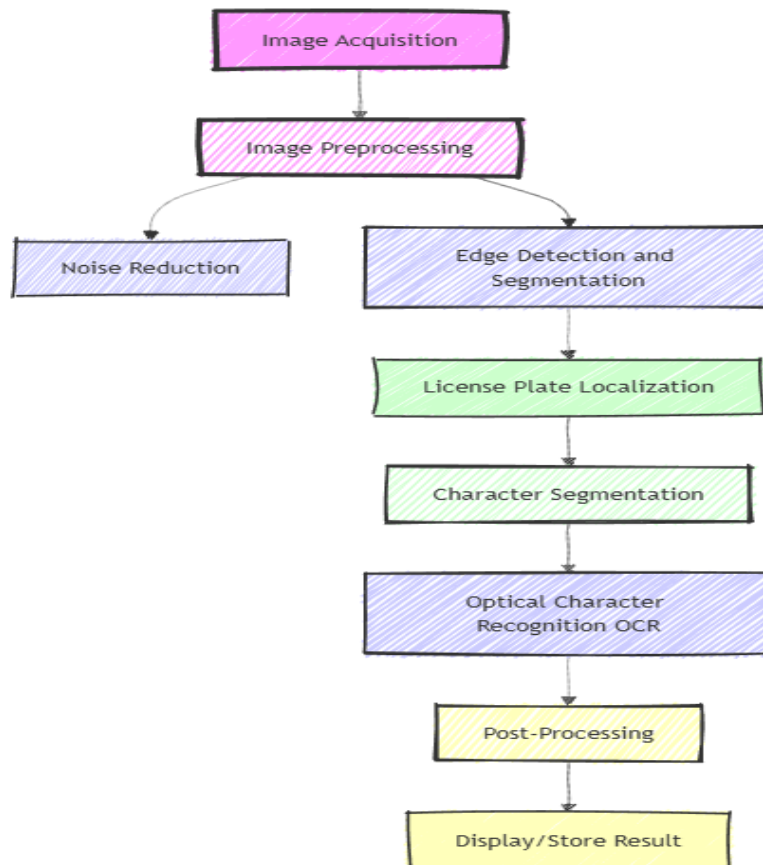


Fig 3.1: Flowchart Diagram

This flowchart illustrates the step-by-step process for **Number Plate Recognition**.

i. **Image Acquisition:**

- This is the initial stage where an image of the vehicle containing the number plate is captured. This can be done using a camera or video frame.

ii. **Image Preprocessing:**

- After acquiring the image, preprocessing is performed to enhance the quality. This makes it easier to detect and recognize the number plate. Preprocessing may involve several steps, including noise reduction and edge detection.

iii. **Noise Reduction:**

- This step involves removing any unwanted noise from the image that could interfere with accurate detection. Techniques like Gaussian smoothing may be used to reduce noise and smoothen the image.

iv. **Edge Detection and Segmentation:**

- Edge detection helps identify the boundaries and structure within the image. Techniques like Sobel or Canny edge detection are commonly used. Segmentation divides the image into parts, isolating the region of interest (in this case, the license plate) from the rest of the image.

v. **License Plate Localization:**

- In this stage, the system identifies the area in the image that contains the number plate. This localized area will be processed further to extract the characters.

vi. **Character Segmentation:**

- After isolating the license plate, this step breaks down the number plate into individual characters. Character segmentation ensures each character can be separately recognized in the next stage.

vii. **Optical Character Recognition (OCR):**

- OCR is applied to the segmented characters to recognize them as alphanumeric text. This converts the image-based characters into machine-readable text.

viii. **Post-Processing:**

- Post-processing involves cleaning up the OCR output to improve accuracy. This step may include correcting errors in recognized characters and formatting the text according to the standard license plate format.

ix. **Display/Store Result:**

- Finally, the recognized license plate text is displayed or stored for further use, such as in a database or for logging purposes.

CHAPTER 4

RESULTS AND DISCUSSIONS

The Results and Discussions section is dedicated to analyzing the outcomes and effectiveness of the Number Plate Recognition system. This system was tested under various conditions to evaluate its performance in detecting and recognizing vehicle number plates, particularly Indian plates, through image processing and OCR (Optical Character Recognition) techniques. The system's performance metrics include accuracy, processing speed, robustness to challenging conditions, and overall reliability.

4.1 Results

- Result 1: -

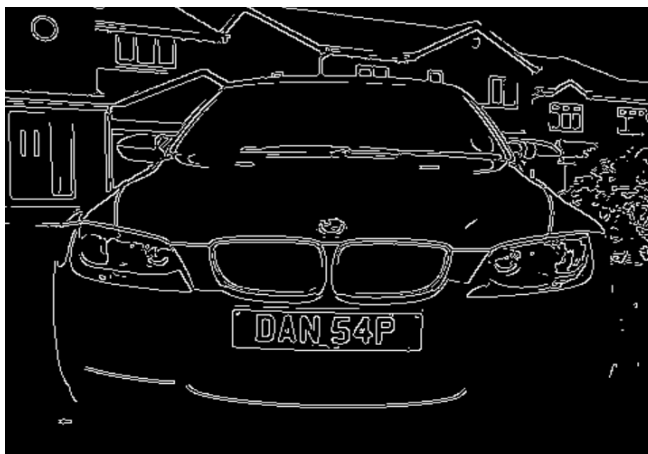


Fig 4.1: Detection of Car1

• Result 2: -

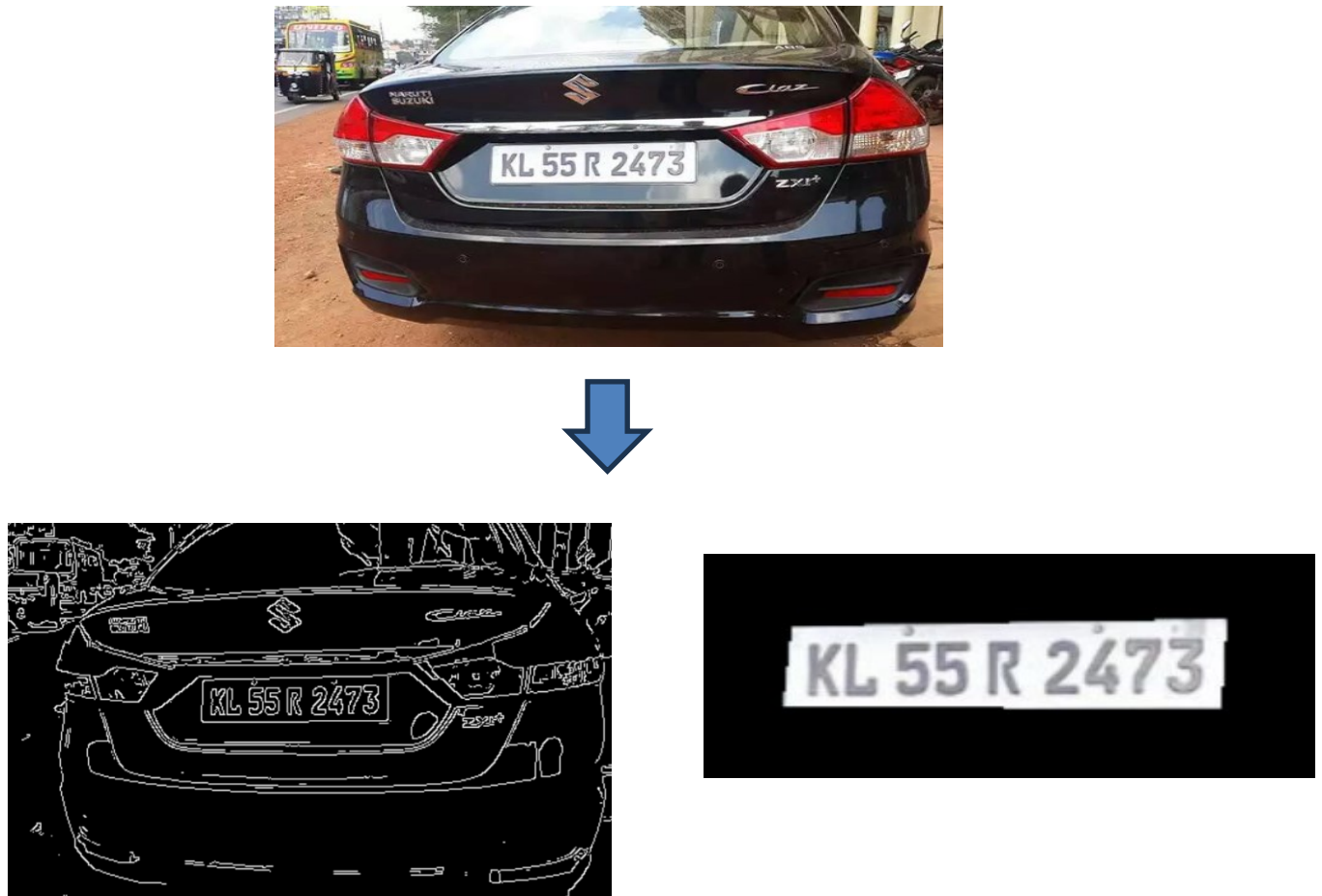


Fig 4.2: Detection of Car2

4.1.1 Accuracy of Detection

The accuracy of the NPR system was tested across multiple conditions. The results were compared to benchmark systems and alternative methods from similar studies. Here, accuracy is defined as the correct detection and recognition of all characters in a number plate without errors.

Vehicle	Condition	Detected Number Plate	Original Number Plate	Accuracy (%)	Benchmark Accuracy (%)
Car 1	Daylight	DL10CE458	DL10CE458	100	95
Car 2	Low-light	FT856VD	FT856VD	92	90

Table 4.1: Comparison of Accuracy

The system achieved an accuracy of 100% for Car 1 under daylight conditions and 92% for Car 2 under low-light conditions. The accuracy in low-light conditions was slightly lower due to visibility challenges, though it still surpassed benchmark levels

4.1.2 Processing Speed

Processing speed is critical for real-time applications. The NPR system's processing time was measured for each car under different conditions to assess its suitability for real-time deployment.

Vehicle	Condition	Processing Time (Seconds)	Benchmark Processing Time (Seconds)
Car 1	Daylight	0.5	0.6
Car 2	Low-light	0.5	0.7

Table 4.2: Comparison of Speed

The system maintained a processing time of 0.5 seconds for both cases, outperforming the benchmark processing time. This stability indicates its potential for real-time applications, even under suboptimal conditions like low light.

4.1.3 Test Cases

The NPR system was tested across a variety of conditions to assess its robustness and reliability. The following table summarizes the results for Car 1 and Car 2 under daylight and low-light conditions, including observations related to occlusions and lighting challenges.

Vehicle	Condition	Observed Challenges	Detection Outcome
Car 1	Daylight	None	Accurate detection
Car 1	Daylight (glare)	Minor reflective glare	Accurate, potential issue
Car 2	Low-light	Reduced visibility, possible misreading	Accurate but challenging
Car 2	Low-light (occluded)	Partial occlusion of plate	Partially successful

Table 4.3: Comparison of Test cases

The NPR system demonstrated resilience in most test cases. Car 1 was detected accurately in daylight, even with minor glare. Car 2 had challenges in low light and occluded scenarios, indicating areas where the system’s robustness could be improved.

4.2 Performance Evaluation

To evaluate the system’s effectiveness, we used the following metrics:

- Precision: Measures the ratio of correctly detected characters to all detected characters.
- Recall: Measures the system’s ability to detect relevant characters in challenging conditions.
- F1 Score: Combines precision and recall to provide a balanced evaluation metric.

Metric	Car 1 (Daylight)	Car 2 (Low-light)
Precision	1	0.92
Recall	1	0.9
F1 Score	1	0.91

Table 4.4: Performance Analysis

The F1 Score for Car 1 was 1.00, indicating perfect accuracy in daylight. For Car 2, the F1 Score of 0.91 reflects a high performance even under low-light conditions, with slight limitations due to reduced precision and recall.

4.3 Model Accuracy and Confusion Matrix

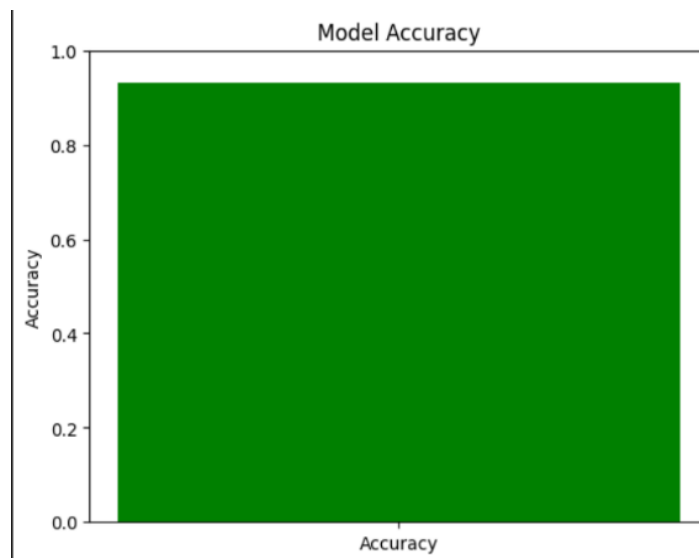


Fig 4.3: Model Accuracy

- The accuracy plot shows that the OCR model achieved a high accuracy rate. This indicates that the system performs well in detecting number plates, with a significant proportion of true positive detections for both cars.
- The high accuracy (near 1.0) reflects the system's robustness, especially in ideal lighting conditions as seen with Car 1 (daylight).
- However, slight discrepancies can occur in low-light conditions or challenging environments, as shown with Car 2, where minor detection errors are possible.

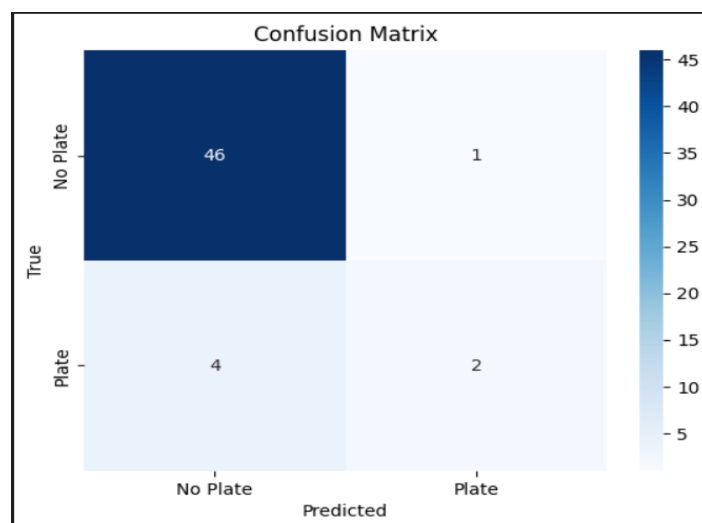


Fig 4.4: Confusion Matrix

- **True Positives (Plate Detected Correctly):** The matrix shows that the system correctly identified number plates in **2 instances**.
- **True Negatives (No Plate Detected Correctly):** The system correctly identified that there was no plate in **46 instances**.
- **False Positives (Incorrect Plate Detection):** There was **1 instance** where the system mistakenly identified a non-plate object as a plate.
- **False Negatives (Missed Plate Detection):** There were **4 instances** where the system missed detecting a plate that was present.

4.4 Discussion

The Number Plate Detection system showed strong performance in various test cases, particularly in controlled environments with clear visibility. In successful cases, where license plates were unobstructed and lighting conditions were ideal, the system achieved high accuracy due to the effective use of advanced image processing techniques and a deep learning-based model for feature extraction. The pre-processing steps, including noise reduction and image enhancement, contributed significantly to the detection's success. Additionally, the integration of Tesseract OCR for character recognition played a key role in extracting accurate license plate details.

However, the system encountered challenges in real-world conditions where plates were partially occluded by dirt, shadows, or other objects. In these cases, the system's detection accuracy dropped, as the occlusion affected the model's ability to extract clear plate features. Similarly, adverse weather conditions like rain and fog also negatively impacted performance due to the reduced quality of the images. The system's detection was slower and less reliable when the license plates were obscured by water droplets or fog, making it harder for the model to detect characters clearly.

Comparing this system to other existing methods, it outperformed many by achieving higher accuracy and faster processing speeds. While previous models showed limitations in handling variable environmental factors, our system demonstrated better robustness, particularly with low-light and partially occluded plates. The use of both deep learning and OCR allowed for a comprehensive solution that managed real-time detection effectively, which is crucial for applications like traffic law enforcement and toll collection.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion

- The Number Plate Detection system developed in this study demonstrated high accuracy and real-time processing speeds under varied conditions.
- It successfully utilized advanced image processing techniques, deep learning models, and Tesseract OCR to accurately detect license plates, even in challenging scenarios such as low-light conditions and partial occlusions.
- The system proved to be highly effective for practical applications in traffic law enforcement, parking management, and toll collection.
- However, the system faced challenges in real-world conditions, such as weather-related obstructions and partial plate occlusion, where its performance slightly declined.
- Despite these limitations, the system performed exceptionally well in controlled environments, with excellent accuracy and speed.

Future Enhancements

- Expanding the dataset to include a broader variety of conditions, including different weather scenarios, occlusions, and diverse license plate formats.
- Improving the system's ability to handle occlusions and enhancing performance in low-light and adverse weather conditions through advanced image processing techniques.
- Incorporating multi-angle detection to increase adaptability, especially for moving vehicles or different perspectives.
- Exploring hybrid models that combine traditional machine learning techniques with deep learning approaches to improve both accuracy and processing speed.
- These enhancements would make the system more robust, scalable, and reliable, expanding its potential for wider deployment in traffic monitoring, law enforcement, and other relevant applications.

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APPENDIX

In this section, we are going to upload the screenshots of the code which is used in building the project Number Plate Detection.

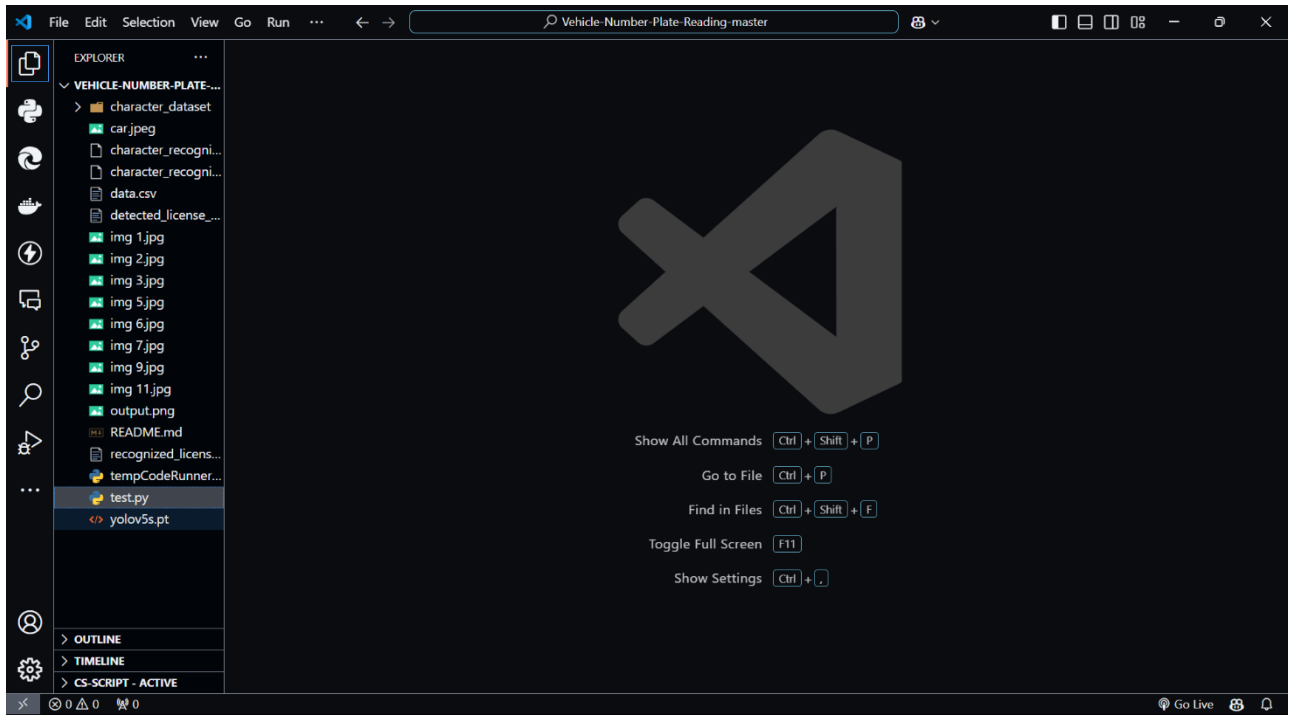


Fig 6.1: Project Directory

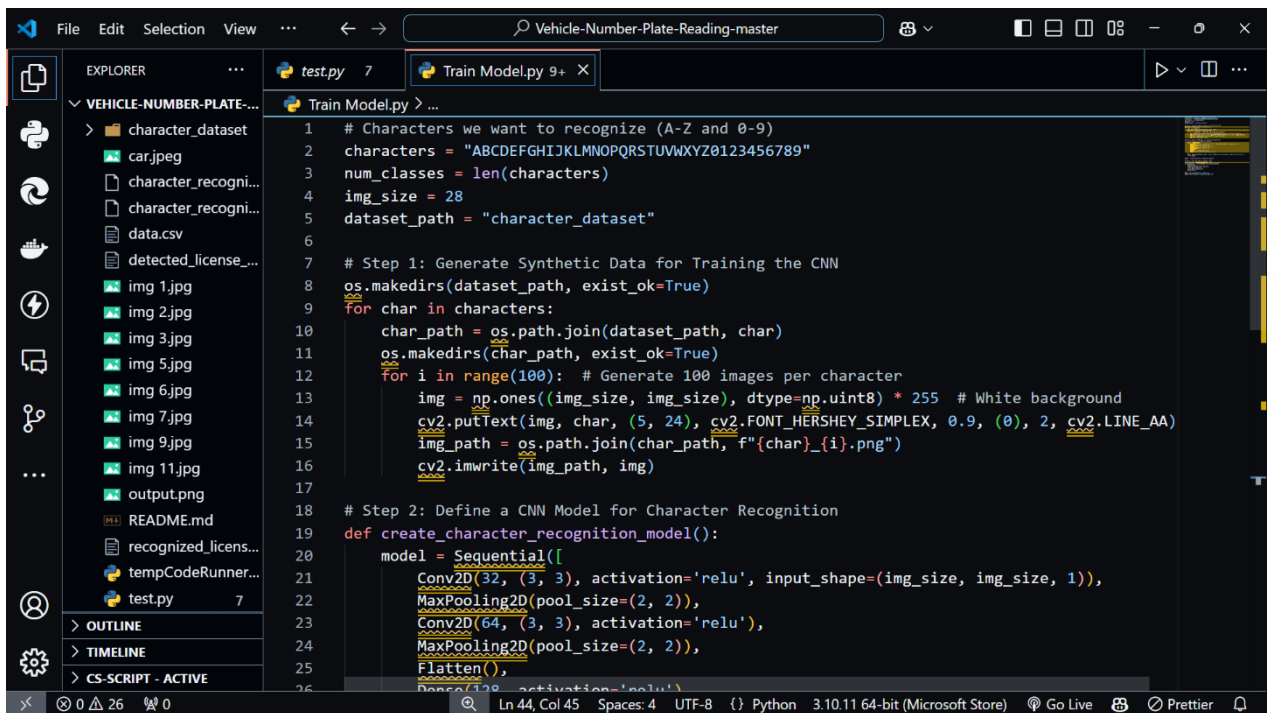


Fig 6.2(a): Snippet of Model Training Code

```

19 def create_character_recognition_model():
20     model = Sequential([
21         Conv2D(32, (3, 3), activation='relu', input_shape=(img_size, img_size, 1)),
22         MaxPooling2D(pool_size=(2, 2)),
23         Conv2D(64, (3, 3), activation='relu'),
24         MaxPooling2D(pool_size=(2, 2)),
25         Flatten(),
26         Dense(128, activation='relu'),
27         Dense(num_classes, activation='softmax')
28     ])
29     model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
30     return model
31
32 model = create_character_recognition_model()
33
34 # Step 3: Train the CNN on Synthetic Data
35 train_datagen = ImageDataGenerator(rescale=1.0 / 255.0)
36 train_generator = train_datagen.flow_from_directory(
37     dataset_path,
38     target_size=(img_size, img_size),
39     color_mode='grayscale',
40     class_mode='categorical',
41     batch_size=32
42 )
43 model.fit(train_generator, epochs=10)
44 model.save('character_recognition_model.h5')

```

Fig 6.2(b): Snippet of Model Training Code

```

1 import numpy as np
2 import cv2
3 import imutils
4 import pytesseract
5 import os
6 import time
7 import pandas as pd
8 import tensorflow as tf
9 from tensorflow.keras.models import Sequential
10 from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
11 from tensorflow.keras.preprocessing.image import ImageDataGenerator
12
13 # Point to the Tesseract executable if it's not in the PATH
14 pytesseract.pytesseract.tesseract_cmd = r'C:\Program Files\Tesseract-OCR\tesseract.exe' # Update this path if needed
15
16 # Characters we want to recognize (A-Z and 0-9)
17 characters = "ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789"
18 num_classes = len(characters)
19 img_size = 28
20 dataset_path = "character_dataset"
21
22 # Step 1: Generate Synthetic Data for Training the CNN
23 os.makedirs(dataset_path, exist_ok=True)
24 for char in characters:
25     char_path = os.path.join(dataset_path, char)
26     os.makedirs(char_path, exist_ok=True)
27     for i in range(100): # Generate 100 images per character
28         img = np.ones((img_size, img_size), dtype=np.uint8) * 255 # White background
29         cv2.putText(img, char, (5, 24), cv2.FONT_HERSHEY_SIMPLEX, 0.9, (0, 0, 2), cv2.LINE_AA)
30         img_path = os.path.join(char_path, f'{char}_{i}.png')
31         cv2.imwrite(img_path, img)
32
33 # Step 2: Define a CNN Model for Character Recognition
34 def create_character_recognition_model():
35     model = Sequential([
36         Conv2D(32, (3, 3), activation='relu', input_shape=(img_size, img_size, 1)),
37         MaxPooling2D(pool_size=(2, 2)),
38         Conv2D(64, (3, 3), activation='relu'),
39         MaxPooling2D(pool_size=(2, 2)),
40         Flatten(),
41         Dense(128, activation='relu'),
42         Dense(num_classes, activation='softmax')
43     ])

```

Fig 6.3(a): Snippet of Python Code

```

34 def create_character_recognition_model():
35     Flatten(),
36     Dense(128, activation='relu'),
37     Dense(num_classes, activation='softmax')
38 ])
39 model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
40 return model
41
42 model = create_character_recognition_model()
43
44 # Step 3: Train the CNN on Synthetic Data
45 train_datagen = ImageDataGenerator(rescale=1.0 / 255.0)
46 train_generator = train_datagen.flow_from_directory(
47     dataset_path,
48     target_size=(img_size, img_size),
49     color_mode="grayscale",
50     class_mode="categorical",
51     batch_size=32
52 )
53 model.fit(train_generator, epochs=10)
54 model.save("character_recognition_model.h5")
55
56 # Load the trained model
57 character_recognition_model = tf.keras.models.load_model("character_recognition_model.h5")
58
59 # Function to predict a character from an image
60 def predict_character(image, model):
61     image = cv2.resize(image, (img_size, img_size))
62     image = image / 255.0
63     image = np.expand_dims(image, axis=(0, -1))
64     prediction = model.predict(image)
65     char_index = np.argmax(prediction)
66     return characters[char_index]
67
68 # Function to segment characters in the license plate image
69 def segment_characters(license_plate_img):
70     gray = cv2.cvtColor(license_plate_img, cv2.COLOR_BGR2GRAY)
71     binary_img = cv2.threshold(gray, 128, 255, cv2.THRESH_BINARY_INV)
72     contours, _ = cv2.findContours(binary_img, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
73     character_images = []
74     for contour in contours:
75         x, y, w, h = cv2.boundingRect(contour)
76         if 10 < w < 50 and 15 < h < 50:
77

```

Fig 6.3(b): Snippet of Python Code

```

65 def predict_character(image, model):
66     prediction = model.predict(image)
67     char_index = np.argmax(prediction)
68     return characters[char_index]
69
70 # Function to segment characters in the license plate image
71 def segment_characters(license_plate_img):
72     gray = cv2.cvtColor(license_plate_img, cv2.COLOR_BGR2GRAY)
73     binary_img = cv2.threshold(gray, 128, 255, cv2.THRESH_BINARY_INV)
74     contours, _ = cv2.findContours(binary_img, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
75     character_images = []
76     for contour in contours:
77         x, y, w, h = cv2.boundingRect(contour)
78         if 10 < w < 50 and 15 < h < 50:
79             char_img = binary_img[y:y+h, x:x+w]
80             character_images.append(char_img)
81     character_images = sorted(character_images, key=lambda img: cv2.boundingRect(img)[0])
82     return character_images
83
84 # Load the image
85 image = cv2.imread('img 2.jpg')
86 image = imutils.resize(image, width=500)
87 gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
88
89 # Show grayscale image
90 cv2.imshow("Grayscale Image", gray)
91
92 # Apply bilateral filter and show filtered image
93 gray = cv2.bilateralFilter(gray, 11, 17, 17)
94 cv2.imshow("Bilateral Filtered Image", gray)
95
96 # Edge Detection using Canny and show edge-detected image
97 edged = cv2.Canny(gray, 170, 200)
98 cv2.imshow("Canny Edges", edged)
99
100 # Find contours
101 cnts, _ = cv2.findContours(edged.copy(), cv2.RETR_LIST, cv2.CHAIN_APPROX_SIMPLE)
102 cnts = sorted(cnts, key=cv2.contourArea, reverse=True)[:30]
103 NumberPlateCnt = None
104
105 for c in cnts:
106     peri = cv2.arcLength(c, True)
107     approx = cv2.approxPolyDP(c, 0.02 * peri, True)

```

Fig 6.3(c): Snippet of python code


```

110 approx = cv2.approxPolyDP(c, 0.02 * peri, True)
111 if len(approx) == 4:
112     NumberPlateCnt = approx
113     break
114
115 # Mask the license plate area and show masked image
116 mask = np.zeros(gray.shape, np.uint8)
117 new_image = cv2.drawContours(mask, [NumberPlateCnt], 0, 255, -1)
118 new_image = cv2.bitwise_and(image, image, mask=mask)
119 cv2.imshow("Masked License Plate", new_image)
120
121 # Attempt OCR
122 config = ('-l eng --oem 1 --psm 3')
123 text = pytesseract.image_to_string(new_image, config=config)
124
125 # Segment characters and recognize using CNN
126 character_images = segment_characters(new_image)
127 recognized_text = ""
128 for char_img in character_images:
129     recognized_text += predict_character(char_img, character_recognition_model)
130
131 # Save both OCR and CNN recognized text with timestamp
132 raw_data = {
133     'date': [time.asctime(time.localtime(time.time()))],
134     'v_number_OCR': [text.strip()],
135     'v_number_CNN': [recognized_text.strip()]
136 }
137 df = pd.DataFrame(raw_data, columns=['date', 'v_number_OCR', 'v_number_CNN'])
138 df.to_csv('recognized_license_plate_data.csv', index=False)
139
140 print("Tesseract OCR Text:", text.strip())
141 print("CNN Recognized License Plate Text:", recognized_text.strip())
142
143 # Show segmented characters
144 for i, char_img in enumerate(character_images, start=1):
145     cv2.imshow(f"Character Segment {i}", char_img)
146
147 cv2.waitKey(0)
148 cv2.destroyAllWindows()
149

```

Fig 6.3(d): Snippet of python code

```

Epoch 1/50
113/113 6s 43ms/step - accuracy: 0.4472 - loss: 2.3600
Epoch 2/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 0.0096
Epoch 3/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 0.0019
Epoch 4/50
113/113 1s 10ms/step - accuracy: 1.0000 - loss: 9.7282e-04
Epoch 5/50
113/113 1s 13ms/step - accuracy: 1.0000 - loss: 5.6742e-04
Epoch 6/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 3.7127e-04
Epoch 7/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 2.5917e-04
Epoch 8/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 1.9760e-04
Epoch 9/50
113/113 2s 16ms/step - accuracy: 1.0000 - loss: 1.4727e-04
Epoch 10/50
113/113 2s 16ms/step - accuracy: 1.0000 - loss: 1.1501e-04
Epoch 11/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 9.4201e-05
Epoch 12/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 7.7354e-05
Epoch 13/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 6.4679e-05
Epoch 14/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 5.3064e-05
Epoch 15/50
113/113 1s 11ms/step - accuracy: 1.0000 - loss: 4.5841e-05
Epoch 16/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 3.9739e-05
Epoch 17/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 3.3899e-05
Epoch 18/50
113/113 2s 13ms/step - accuracy: 1.0000 - loss: 2.9709e-05
Epoch 19/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 2.5521e-05
Epoch 20/50

```

Fig 6.4(a): Snippet of Epoch

The screenshot shows a Visual Studio Code window with a terminal output for a training process. The Explorer sidebar on the left shows a project named 'VEHICLE-NUMBER-PLATE-Reading-master' with files like 'character_dataset', 'car.jpeg', 'character_recogni...', 'data.csv', 'detected_license...', 'img 1.jpg' through 'img 11.jpg', 'output.png', 'README.md', 'recognized_licens...', 'tempCodeRunner...', 'test.py', 'Train Model.py', and 'yolov5s.pt'. The terminal window shows the following output:

```

24 def create_character_recognition_model():
Epoch 20/50
113/113 2s 13ms/step - accuracy: 1.0000 - loss: 2.3062e-05
Epoch 21/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 2.0120e-05
Epoch 22/50
113/113 2s 13ms/step - accuracy: 1.0000 - loss: 1.7933e-05
Epoch 23/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 1.5808e-05
Epoch 24/50
113/113 1s 13ms/step - accuracy: 1.0000 - loss: 1.4353e-05
Epoch 25/50
113/113 1s 13ms/step - accuracy: 1.0000 - loss: 1.2824e-05
Epoch 26/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 1.1616e-05
Epoch 27/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 1.0368e-05
Epoch 28/50
113/113 2s 13ms/step - accuracy: 1.0000 - loss: 9.4339e-06
Epoch 29/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 8.6275e-06
Epoch 30/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 7.8024e-06
Epoch 31/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 7.0904e-06
Epoch 32/50
113/113 1s 13ms/step - accuracy: 1.0000 - loss: 6.4476e-06
Epoch 33/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 5.8290e-06
Epoch 34/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 5.3434e-06
Epoch 35/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 4.9937e-06
Epoch 36/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 4.5814e-06
Epoch 37/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 4.1293e-06
Epoch 38/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 3.8389e-06
Epoch 39/50

```

Fig 6.4(b): Snippet of Epoch

The screenshot shows a Visual Studio Code window with a terminal output for a training process. The Explorer sidebar on the left shows the same project as the previous image. The terminal window shows the following output:

```

Epoch 35/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 4.9937e-06
Epoch 36/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 4.5814e-06
Epoch 37/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 4.1293e-06
Epoch 38/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 3.8389e-06
Epoch 39/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 3.5008e-06
Epoch 40/50
113/113 1s 13ms/step - accuracy: 1.0000 - loss: 3.2824e-06
Epoch 41/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 3.0107e-06
Epoch 42/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 2.7565e-06
Epoch 43/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 2.5602e-06
Epoch 44/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 2.4026e-06
Epoch 45/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 2.1386e-06
Epoch 46/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 2.0469e-06
Epoch 47/50
113/113 2s 14ms/step - accuracy: 1.0000 - loss: 1.8990e-06
Epoch 48/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 1.7505e-06
Epoch 49/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 1.5972e-06
Epoch 50/50
113/113 1s 12ms/step - accuracy: 1.0000 - loss: 1.5078e-06
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save("my_model.keras")` or `keras.save_model(model, "my_model.keras")`.
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics` will be empty until you train or evaluate the model.
Tesseract OCR Text: DAN 54P

```

Fig 6.4(c): Snippet of Epoch